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Systematic review of rural drinking water interventions under Jal Jeevan mission

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Abstract

Access to clean and safe drinking water is a fundamental human right and a cornerstone of sustainable development. Among all the Sustainable Development Goals, SDG 6 (Clean Drinking Water and Sanitation) is distinct, as it aims to ensure proper, clean and safe drinking water and sanitation for all. To achieve SDG 6, the Government of India has launched the Jal Jeevan Mission (JJM) scheme. There are numerous problems and challenges associated with implementing the scheme. It involves various challenges and problems like water scarcity, inequitable distribution of water resources, lack of infrastructure, cultural belief systems and geographical locations. Despite a lot of challenges, Jal Jeevan Mission FHTC was successfully installed in many parts of the country. The study employs Systematic Literature Review (SLR) through the PRISMA [1] method to identify how the social, political and financial constraints influence the access to water. Thematic analysis is performed to understand how socio-economic benefits among the rural community and how it promotes gender equity. The findings also elaborates the transformation in public health and sanitation outcomes and advocates the importance of Community Participation, and Local Governance integration in developing a sustainable drinking water management.

Keywords: JJM, SDG 6, Community participation, Drinking water, Sustainability, Policy

1 Introduction

In 2015, the United Nations (UN) adopted the 17 Sustainable Development Goals (SDGs) to address the global challenges by 2030 widely focusing on poverty, equality, education, health and environment sustainability. The study explicitly focuses on Sustainable Development Goal 6, i.e. clean and safe drinking water for all. To attain the goal, as a national initiative, the Government of India launched a flagship programme, Jal Jeevan Mission (JJM), which aims to install Functional Household Tap connections (FHTCs) to all the households in the country by 2028. In varied landscape and mixed terrain like India, achieving the JJM involves copious practical difficulties such as lack of infrastructure, water scarcity, and cultural belief systems which demand large scale interventions. The multidimensional challenges are addressed by the Mission by adopting a community-centric approach.

The study carefully examines the selected articles to explore the JJM (implementation, problems, challenges and ways to achieve equitable drinking water supply) [2].



JJM makes significant efforts to maintain the quality of water and it also takes measures to monitor the water quality in all regions. Analytical and empirical approach of JJM is necessary to overcome the operational and infrastructure limitations. Through this approach, multidimensional perspectives (such as social inclusion, financial accountability, environmental stewardship, local governance and long-term resilience) are analysed to improve the Mission's progress. Although JJM has several backdrops, the systematic review of literature provides evidence on the impact and challenges of JJM, thereby contributing to global rural water governance discourse.

Though expanding literature has investigated rural drinking water access, public health outcomes and water resilience in India, existing studies on JJM still remain fragmented and majorly issue-specific. Most of the prior studies focus on water quality, service delivery, gender impacts or case-studies, there is a clear lack of evidence across various factors like socio-economic, health, governance and sustainability dimensions to evaluate the overall JJM outcomes.

Additionally, less focus has been given to JJM within broader global water governance and SDG 6 discourses. To address this gap, a Systematic Review of Literature (SLR) is undertaken to consolidate the existing evidence and to identify the recurring patterns and gaps. The study also critically analyses JJM's contribution and challenges in advancing rural drinking water resilience. The findings of study provide insights to future policy makers, and comparative water governance research. The SLR comprehensively synthesizes the growing body of literature on JJM. SLR through its rigorous approach provides consolidated evidence on common patterns, issues, consistencies and divergences in the study findings.

Unlike empirical or narrative studies, SLR offers transparent, reproducible and unbiased synthesis of published scholarly research works. It follows a structured search process, screening and analysis. The SLR approach is appropriate to evaluable socio-economic, health, governance and sustainability dimensions of JJM in an integrated manner. Through the SLR approach, the study provides robust evidence to policy formulation, direction for future researchers, and effective water governance strategies that directly align with SDGs 6.

2 Methodology

In this study, to comprehend and to interpret the challenges of the Jal Jeevan Mission, scholarly articles were collected and Systematic Review of Literature (SLR) procedure was employed [3]. Through Preferred Reporting Items for Systematic Reviews and Meta Analyses [1] technique the data are curated to extract recurring themes. The scholarly articles were collected from Google Scholar, Sage, Researchgate, Scopus and Elsevier. Key words such as 'Jal Jeevan Mission', 'JJM', 'Drinking water programs in India', 'Challenges in JJM', 'Rural water supply in India' are used to gather data from the database. Through PRISMA 2020[4] technique, findings of the studies are consolidated, which helps the researchers to identify the process of implementation, challenges in Jal Jeevan Mission, requirement to achieve JJMs goal. Google Spreadsheet and NVivo12 softwares were used to curate and analyse the data collected. After curation, thematic analysis was performed to identify the recurring themes.

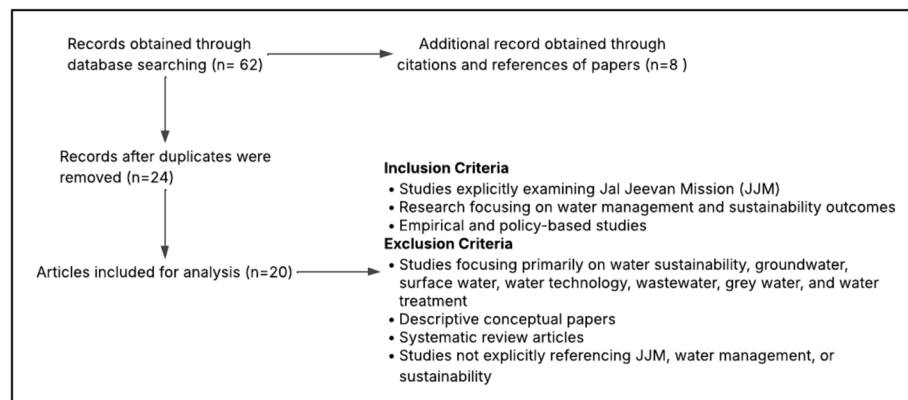


Fig. 1 PRISMA 2020 flow diagram: Study selection process

2.1 Formulating the problem

The study aims to review the research work conducted on Jal Jeevan Mission (JJM) as a policy instrument to achieve equitable water access. The formulated specific objectives are:

- To identify the research gaps in JJM as a policy instrument to achieve equitable water access.
- To understand the socio-economic contexts that shape water service delivery
- To evaluate Jal Jeevan Mission's alignment with Sustainable Development Goals.

2.2 Developing and validating the review process

To explore the socio-economic context, existing disparities and local water governance that shapes the water service delivery and to evaluate the Jal Jeevan Mission's alignment with Sustainable Development Goals, systematic review analysis was adopted [3] and the data are presented in Fig.1. Word Cloud was created using NVivo12 software. Initially relevant literature was collected, then screened for inclusion. The quality of the screened articles were analysed and synthesised. The findings were reported after performing the analyses.

3 Searching the literature

3.1 Study selection and screening process

The systematic review followed the PRISMA 2020 guidelines. Records were identified through database searches in Google Scholar, Scopus, ScienceDirect, Sage, Taylor & Francis, DOAJ, Elsevier, and ResearchGate. All retrieved records were exported to a spreadsheet, and duplicate entries were identified and removed.

The remaining records were screened independently by the authors based on titles and abstracts using predefined inclusion and exclusion criteria. Studies that appeared relevant were subjected to full-text assessment for eligibility. Articles were excluded at the full-text stage if they did not focus on the Jal Jeevan Mission, lacked empirical

or analytical content, were not available in full text, or were not published in English. Any disagreements during the screening process were resolved through discussion among the authors.

The complete study selection process, including reasons for exclusion at each stage, is presented in the PRISMA flow diagram (Fig. 1).

3.2 Backward and forward search

The study adopted backward and forward search to explore the important research articles cited in the selected research papers and to find the articles that cited the particular selected research paper. Google scholar database facilitates the search with its more comprehensive search capabilities [5]. For this study there were 8 papers found via backward and forward search.

3.3 Keywords used for the search

The selected literature articles focusing on Jal Jeevan Mission, importance of clean and safe drinking water, Government initiatives to provide drinking water accessibility to various regions and policies were selected as the sample for this study. In each database, the search terms entered were, 'Jal Jeevan Mission or JJM', 'Drinking water programs in India', 'Rural water supply systems in India', 'implementation challenges in Jal Jeevan Mission'.

3.4 Sampling strategy

The research study enfold comprehensive review of literature such as scholarly articles, reviews papers and book chapters. The study adopted PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) 2020 technique. The sampling strategy is presented in Fig. 1. The sample includes published articles on various areas like management studies, water resource management and conservation, grey water management, sustainable water practices, community engagement, Government policies, business communication, Government initiatives and health communication.

The sampling includes articles published between January 2002 and May 2025. The researcher analyses the relevant literature from 2000 to understand the accessibility of clean, safe and adequate drinking water in rural India. May 2025 was the exit point of data collection. Only English language studies were selected for the review and analysis.

3.5 Refining results with additional restrictions

A total of 62 records were initially identified through database searching. After removing duplicate entries, 24 unique studies remained and were subjected to title and abstract screening. Following full-text assessment for eligibility, 4 articles were excluded based on the predefined inclusion and exclusion criteria. Finally, 20 articles met all eligibility requirements and were selected for detailed qualitative analysis.

3.6 Stopping rule

In the study, for the Google scholar database, the search was limited to 20 articles as there were no relevant articles found after that.

3.7 Inclusion and exclusion criteria

Inclusion and exclusion criteria were applied at two stages. During the title and abstract screening stage, studies were excluded if they were unrelated to rural drinking water interventions or the Jal Jeevan Mission. During the full-text screening stage, studies were excluded if they lacked relevance to policy implementation, outcomes, or challenges of the Jal Jeevan Mission, did not provide empirical or analytical insights, or were not accessible as full-text publications. This two-stage screening ensured the relevance and quality of the final set of studies included for synthesis.

3.8 Screening procedure

The selected article's title, abstract and the keywords were screened to match the inclusion/exclusion criteria.

3.9 Criteria for quality assessment and procedures

After selecting the sample for the study based on the screening procedures of inclusion/exclusion criteria, the articles with full-text were collected and read thoroughly. The journal articles, books and book chapters published by reliable and reputed scholarly journals and books were included in the study. The quality assessment work was parallelly assigned to all the three authors of the review. Discrepancies in the findings of the study were discussed and sorted.

4 Extracting data for analysis

The Fig. 1 illustrates the systematic process followed for identifying, screening, and selecting studies for this review. In the identification stage, a comprehensive literature search was conducted and 62 records were initially identified. In addition, 8 relevant studies were identified through backward and forward citation searching, resulting in a total of 70 records.

All identified records were compiled into a single spreadsheet, and duplicate entries were carefully removed. After removing the duplicate records, 24 records remained for screening. During the screening stage, these 24 records were screened based on their titles, abstracts and predefined inclusion and exclusion criteria. During this screening stage, 4 studies which did not directly study JJM or rural drinking water interventions/policies, studies which lack empirical or analytical relevance and absence of full-test were excluded.

After the screening process, finally 20 studies were included for qualitative synthesis and formed the basis of thematic analysis. Through thematic analysis, key themes such as socio-economic impacts, public health outcomes, challenges and community-participation under JJM were identified. The PRISMA flow diagram (Fig. 1)

demonstrated the transparent, reliable, and systematic review process ensuring methodological rigour.

4.1 Data synthesis

To systematically organise, code and synthesise qualitative evidence from the selected articles, thematic analysis was performed using NVivo 12 software. The final 20 articles after curation were imported by NVivo for analysis. Through an inductive-deductive approach, review objectives and recurring concepts, the initial coding framework was developed.

Initially, open coding was performed to identify units related to implementation of policy, community participation, socio-economic impacts, public health outcomes and infrastructural challenges under JJM. The codes were then refined and grouped into broader categories. The themes were consolidated into higher-order themes to identify patterns across studies. In order to ensure rigour, the coding process was reviewed by all the authors. Consistency across the sources were verified using NVivo's query and node comparison functions. The broader and final themes were validated and aligned. The validation and coding process ensured reliability, reproducibility and transparency of the thematic analysis.

5 Findings

The results were presented based on the distribution of the article publication year, journal, author name, research approach, research method, sampling and sample size. Recurring themes, findings, challenges, and recommendations related to the Jal Jeevan Mission were identified and grouped under relevant categories using Thematic analysis. The study followed a transparent and structured review process. It ensures coherence and clarity in the identification and analysis of relevant literature.

5.1 Distribution of articles by the journal

The selected articles were subjected to detailed analysis from 19 different sources. They are: The Journal of Developing Areas, International Journal of Research and Analytical Reviews, Journal of Emerging Technologies and Innovative Research, Journal of Water, Sanitation and Hygiene for Development, International Journal of Sustainable Development & World Ecology, Human Rights and Social Responsibility: A Reflection, Sustainability, *Frontiers in water (Google scholar)*, Public Health, Journal of Hydrology (*Science Direct*), International Journal of Rural Management (*Sage*), International Journal of Innovative Research in Technology and Science, Journal of Research Administration, International Journal of Enhanced Research in Educational Development, Journal of Global Health Reports, Economic Growth and Sustainable Development: Emerging Trends (*Researchgate*), PloS one, International Journal of Environmental Sciences, (*Scopus*), and Journal of Environmental Management (*Elsevier*) (Table 1).

5.2 Nature of the study

Most of the articles used Qualitative approach (11 articles) research in the selected article, followed by Quantitative approach (6 articles) and Mixed method approach (3 articles). Refer Table 2.

Table 1 List of selected articles for analyses

S. No.	Author	Title
1	Abimash Singh and Gopal Naik [6]	Rural drinking water supply program and societal development: Evidence from the early implementation phase of India's Jal Jeevan Mission
2	Vibha Joshi and Arun Patel [7]	A Review of Har Ghar Nal Yojana (Jal Jeevan Mission): A Case Study of 10 Villages
3	Patil et al. [8]	Potential Health and Economic Gains form Jal Jeevan Mission: A Comprehensive Analysis
4	Ranjeet Kumar [9]	Impact Assessment of the Jal Jeevan Mission in Rural Jharkhand: An Analysis of Tap Water
5	Chaudhuri et al. [10]	Water for all (har ghar jal): rural water supply services (RWSS) in India (2013–2018), challenges and opportunities
6	Sheel et al. [11]	Water as a social determinant of health: bringing policies into
7	Chitra and Pradeepan [12]	Current Scenario of Dynamic Water Resources in Tamil Nadu
8	Agesa and Agesa [13]	Time spent on household chores (fetching water) and the alternatives forgone for women in Sub-Saharan Africa: Evidence from Kenya
9	Ahmed et al. [14]	Fulfilling Domestic Water Demand in Semiarid Regions of North Karnataka: Challenges and Way Forward
10	Kumar & Goyal [9]	Water policy review: Ensuring sustainable water management for India
11	Ahankari et al. [15]	Access to a piped water supply is positively associated with haemoglobin levels in females living in rural Maharashtra State, India
12	Bhuyan and Deka [16]	Investigating the geographies of water use and water cover in the rural areas of Nagaon District, Assam (India): Unraveling the micro-ecological implications
13	Kalimuthu and Monika [17]	Evaluating The Effectiveness OfThe Jal Jeevan Mission In Ensuring Sustainable And Affordable Drinking Water In The Select Taluks Of Tiruppur City, Tamilnadu
14	Basu et al. [18]	Drinking Water: A Myth or Reality in the Indian Subcontinent?
15	Sahoo et al. [19]	India's Prospects for Attaining Sustainable Development Goals on Health & Sanitation: A Critical Analysis of Swachh Bharat Abhiyan & Jal Jeevan Mission
16	Sarkar and Bharat [20]	Achieving Sustainable Development Goals in water and sanitation sectors in India
17	Khalid et al. [21]	Concerns of developing countries and the Sustainable Development Goals: Case for India
18	Brar [22]	The Issues and Concerns related to Water and Sanitation: The Role of Local Government
19	Wescoat Jr, J. L., & Murty, J. V. [23]	District drinking water planning for sustainability in Maharashtra: between local and global scales
20	Wescoat Jr, J. L., Murty, J. V., Singh, R., & Verma, P. [24]	A sustainability planning framework and methods for rural drinking water in Satara District, Maharashtra, India

Table 2 Research methodology

Research Methodology	Frequency of the articles
Quantitative approach	6
Qualitative approach	11
Mixed method approach	3

5.3 Research method

Most of the articles used Survey method (7 articles) research in the selected article, followed by Interview method (6 articles), Case study (4 articles), Descriptive analysis

Table 3 Research method

Research Method	Frequency of the articles
Interview	6
Case-study	4
Survey	7
Content analysis	2
Descriptive analysis	4

method (4 articles) and Content analysis (2 articles). *Refer Table 3.*

6 Thematic analysis and discussion

The analysis of the selected 20 scholarly articles have revealed four primary themes of emphasis. First, the articles highlight how the Jal Jeevan Mission has socio-economic benefits among the rural community and how it promotes gender equity. The second theme, how access to clean and safe drinking water has improved the rural communities health and well-being. It elaborates the transformation in public health and sanitation outcomes. The third theme highlights the challenges related to the implementation of the Jal Jeevan Mission, like Infrastructure set up issues, governance, and service delivery challenges. The fourth theme, advocates the importance of Community Participation, and Local Governance integration in developing a sustainable drinking water management.

6.1 Theme 1: socio-economic empowerment and gender equity

One of the most prominent themes emerging from the literature review is the socio-economic transformation occurred by the Jal Jeevan Mission, specifically focussed on gender equity. Singh and Naik [6], Kumar [9] and Patil et al. [8] explained the implementation of piped water supply which significantly reduces the daily hours spent on water collection. This has enabled women and girls to reallocate their time to income-generating activities, pursuing education, child-rearing—thus sequentially increasing economic productivity and well-being. The findings of the study strongly support the rationale behind JJM as a gender-transformative strategy.

In India, the relief of physical labour and convenient access to the water afforded by the JJM contributed majorly in enhancing the dignity and safety for women. The access to water to Functional Household Tap Connection improved the health and nutrition outcomes among the beneficiaries. Ahankari et al. [15] findings found that access to piped water and higher hemoglobin levels are directly associated. Access to water directly contributes to better nutritional levels and reduced physical exertion.

The comprehensive effects of these changes are far-reaching. Infrastructural limitations and operational disparities in the Jal Jeevan Mission limit the gains for the marginalised section of the society [9, 18]. To enable equitable access to water, further attention must be given to underserved and marginalised areas and attain high-quality water access. Khalid et al. [21] illustrates the importance of embedding equity and justice in

policy design. JJM is often noted as a high-potential initiative, but it requires stronger social audits, bottom-up planning and climate resilient infrastructure.

6.2 Theme 2: public health and sanitation outcomes

The JJM has multifaceted impact on the rural public health and sanitation outcomes. JJM played significant role in reducing the impact of water-borne diseases. Patil et al. [8] and Sheel et al. [11] findings provided empirical evidence on JJM implementation and how drastically it has reduced the impact of water-borne diseases. JJM alleviates the financial pressure on rural India households and allows greater investment in education, nutrition and health-promoting behaviours.

The provision of JJM benefits increased the maternal and child health among the rural households in India. Sheel et al. [11] highlights the water as a critical social determinant of public health and hygiene. The cross-sectional integration of water supply programs and public health interventions emphasises the reduction in parasitic infection and anemia among women and children. The implications of JJM are highly relevant for states with high maternal and child mortality rates and child malnutrition rates. Sahoo et al. [19] and Basu et al. [18] illustrates the importance of ensuring continuous and consistent water safety, by installing water monitoring systems, quality assurance systems, and checking the risk of pathogen exposure levels at regular intervals. Lako and Çomo [25] highlight the importance of developing strategies and policies that focus on wastewater treatment, environmental impact assessment, and modernisation of water supply systems. The findings advocate for an integrated approach to address wastewater treatment challenges, ensuring future water management initiatives are informed by comprehensive evidence and understanding of their environmental impacts.

Sheel et al. [11] also highlights the importance of training community health workers to maintain and monitor water hygiene. The study also argues that the institutionalisation of all the key planners like regional-national levels of health authorities, education sector and Department of drinking water is essential. This cross-sectoral association enables continuous maintenance in water quality and service delivery. Jal Jeevan Mission's contribution to the health sector is underexplored.

6.3 Theme 3: infrastructure development and service delivery challenges

The JJM framework predominantly focuses on infrastructure development and efficient service delivery. Despite the significant progress of the JJM, there are several regional disparities in terms of quality, consistency and sustainability of water supply [6, 19]. Sahoo et al. [19] highlights how the rural regions often face challenges in establishing the infrastructure. The limitation often forced people to rely on unsafe drinking water resources. Singh and Naik [6] mentions how technical failures such as faulty storage tanks and leakages in underground water pipelines are often ignored by the bureaucratic mechanisms. Joshi and Patel [7] also ascertained in the study that infrastructural delays, irregular water pressure, lack of proper maintenance are persistent concerns among the villagers.

Kumar [9] mentions the lack of properly trained operators, the ambiguity over responsibility (between the local Government institutions, Jal Jeevan Mission workers and contractors) also leads to poor maintenance, delayed responses and community

dissatisfaction. The study also emphasises the absence of structured grievance redressal mechanisms in the Jal Jeevan Mission which aggravates the existing problems and challenges of the people in accessing the water. Chaudhuri et al. (2020) findings also mentioned fragmented service delivery, inadequate funding, poor maintenance, climate variability, lack of skilled labour and institutional overlap as major constraints. Additionally, complex water resource dynamics were explored by Chitra and Pradeepan [12]. The findings state that over-reliance on groundwater, erratic monsoons and insufficient recharge systems acts as a barrier in creating sustainable piped water delivery systems. Basu et al. [18] argues the need for involving community participation, need based planning and local hydrogeological realities. This approach leads to developing a sustainable and long term infrastructure and service.

To overcome the challenges, scholars propose a decentralised and context-sensitive infrastructure model, which includes village-level technicians, allocating dedicated budget for operation and management, incorporating digital tools and real-time monitoring systems to avoid service and service disruption. Only through addressing the gap in design, operation and maintenance, the long-term goals of health, equity and empowerment can be achieved.

6.4 Theme 4: community participation and local governance

The active community participation and local governance plays a vital role in the success and sustainability of the JJM. The policy framework of the Jal Jeevan Mission emphasises on the involvement of the local community and the Gram Panchayats in the planning and implementation of the JJM Singh and Naik [6] and Basu et al. [18] study suggest that active local participation and involvement remains limited in many rural areas. Most of the decision making is done by the external engineers or contractors, rather than the local stakeholders. The community participation involves empowerment of the locals in making an informed decision. Kumar [9] and Brar [22] findings states that wherever Local Panchayats are effectively involved in the JJM, mobilising labour, managing local contributions and infrastructure maintenance are well managed. The Panchayat's involvement acts as a local grievance redressal platform.

In many areas, often the marginalized section of the society and women are neglected or underrepresented in the decision making process. Basu et al. [18] emphasises on inclusivity and transparency to overcome the existing inequalities in the governance of rural water management systems. Capacity building programs are crucial in active participation of the local stakeholders. Sahoo et al. [19] highlights the importance of training the local stakeholders in technical, financial management and right-based approach to water governance. Patil et al. [8] mentions the integration of a digital dashboard with local monitoring committees to improve transparency and accountability. Ahmed et al. [14] stresses the requirement of a water security system, decentralised management and community-level rain water harvesting system. Bhuyan and Deka [16] stresses the importance of community water literacy and local water governance in enhancing water resilience.

Wescoat and Murty [23] proposes that a decentralised and data-informed planning approach helps in bridging the existing gap between the national targets and ground-level needs, ultimately making the Jal Jeevan Mission more adaptive and equitable in

Likewise, India's flagship JJM has time-bound targets to increase the household-level water tap connections. However, evidence from global water access programmes such as rural water supply reforms in South Africa [27], decentralised water committees in Kenya [28], water policy measures of Bangladesh [29] and participatory water governance models in Brazil [30] suggests that sustainable water outcomes is achieved through community ownership, tariff rationalisation, and continuous infrastructure and technical support from the government. The review of selected studies indicate that challenges such as intermittent supply, continuous maintenance of infrastructure and inequitable service delivery are observed in the Indian context. The findings highlight the stronger post-installation governance mechanisms for long-term success of the programme.

When analysed through the lens of policy perspective, the scholarly works highlight that in order to achieve climate-resilient water management systems, integrating public health and gender-specific factors are very important. Global evidence also posits that maximising health and hygiene benefits, improving socio-economic conditions and modifying behavioural changes could manifold the outcomes of water governance initiatives. On the contrary, the impact of JJM remains underexplored in terms of nutritional security, maternal health and climate vulnerability.

The global comparisons emphasize JJM as a transformative policy intervention, while its long-term success depends on embedding successful global practices like infrastructure delivery, community-participation, financial support from the government and continuous post- installation monitoring of the policy. The study findings revealed notable regional variations in the implementation of JJM and the expected outcomes. The reasons for the regional disparities are local hydrogeological conditions, local governance, financial support, and communication engagement. In order to overcome the challenges, region-specific modifications and flexible strategies have to be incorporated under JJM to ensure equitable and effective policy outcomes across India.

7 Conclusion

The Jal Jeevan Mission is a pivotal intervention of India's pursuit to improve the accessibility of drinking water to rural population. The JJM has profound impacts on the socio-economic factors, health and hygiene, and local governance. Jal Jeevan Mission had challenges in terms of implementation in few regions, infrastructure quality issues, and lack of administrative efficiency. Jal Jeevan Mission has a transformative potential, specifically it improved the rural livelihoods, public health, and community engagement.

One of the prominent impacts created by the JJM is its influencing and improvising the socio-economic conditions in rural India. Singh and Naik [6] and Sultana et al. [31] highlights how easy access to water reduces the time and effort of women, and how the FHTC indirectly empowers women in the community. Naik et al. [32] mentioned the employment generation potential in the process of infrastructure development and maintenance of JJM. This would potentially stimulate the nation's economic growth contributing to the GDP. Universal access to clean and safe drinking water has several notable health improvements like reduction in waterborne diseases in the rural community. Additionally the regional disparities related to the implementation of the JJM were a recurring theme. Some districts, particularly in drought-prone areas like Raichur and Ballari in Karnataka, and some parts of Jharkhand still continue to struggle due to the

lack of surface water infrastructure and the heavy reliance on groundwater [14]. Regional customization of water supply solutions and increased investment in infrastructure are required to cover the issues related to varied geographical locations.

In conclusion, the JJM has made substantial strides in enhancing the lives of rural populations by providing universal access to clean and safe drinking water, improving health and hygiene, and stimulating economic growth. More focused and customized efforts are required to bridge the regional disparities, particularly in drought-prone and remote areas, where water infrastructure is still lacking.

7.1 Policy implications and future water governance strategies

The findings of this systematic review provide actionable evidence that can directly inform both policy formulation and on-ground implementation of rural drinking water programmes. By synthesising recurring challenges and enabling factors across multiple studies, the review identifies priority intervention areas where policymakers can refine programme design, allocate resources more effectively, and strengthen institutional mechanisms. Rather than offering broad normative recommendations, the evidence generated through this review supports targeted decision-making related to service sustainability, governance accountability, and inter-sectoral coordination within the Jal Jeevan Mission framework.

The findings of this systematic review offer clear policy-relevant insights aligned with the objectives of Sustainable Development Goal 6. First, the evidence underscores the need to shift policy focus from infrastructure coverage alone towards ensuring the sustainability, reliability, and quality of household water supply. Post-installation monitoring mechanisms, routine water quality testing, and dedicated operation and maintenance funding should be institutionalised as core components of JJM.

The review also highlights the importance of strengthening decentralised water governance by empowering Gram Panchayats and local water committees with technical capacity, financial autonomy, and clearly defined accountability mechanisms. By integrating drinking water interventions with public health, sanitation, and nutrition programmes emerges as a critical strategy to maximise social and health returns. The water governance strategies must incorporate climate-resilient and region-specific planning approaches that account for hydrogeological variability, water scarcity, and climate risks.

8 Limitations of the study

While this systematic review provides a comprehensive synthesis of existing literature on the Jal Jeevan Mission, certain limitations should be acknowledged. Mainly the review is limited to studies published in English, and excluded the regional-language publications and grey literature. Further, the study only used secondary data, and does not incorporate field-level evidence or beneficiary perspectives. The study did not capture real-time implementation dynamics. Despite the limitation, the findings offer valuable insights to the policy makers and future researchers about JJM.

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Author contributions

All authors contributed to the study conception and design. Material preparation such as drafting the Introduction and Review of Literature was contributed by K. Ilangovan. Data collection and analysis were performed by Malini Srinivasan.

Jeganathan Gomathi Sankar contributed to the Data analysis and drafting the manuscript. All authors read and approved the final manuscript.

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Data availability

All data supporting this systematic review are cited within the manuscript.

Declarations

Ethics approval and consent to participate

Not applicable as no primary data was collected in the study.

Consent for publication

Not applicable as there are no primary data collected for the study. No human participants are involved in the study.

Competing interests

The authors declare no competing interests.

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